

Microorganisms

Have you ever thought how milk gets sour if it is left out in the open during hot weather? Why do vegetables and fruits rot? Why does garbage stink? Why do we catch the flu? It is because of the germs. Do you know that a lot of really tiny organisms, like germs are present in our environment? They are everywhere and they affect us in various ways. As you have studied that we need to wash our hands with soap and water, before and after eating, to protect ourselves from these germs. The germs might enter our body via dirty hands, if they are not washed properly. Do you know that there are many groups of these tiny organisms?



Figure 2.1: A picture of dirty finger-nails

What gets under your nails? Is it only dirt? No, there are thousands of tiny germs that can make you seriously ill. What should we do to stay safe from germs? Why should you trim your nails?

Unscramble the
given word:
oomcirragnsim

HINT: very tiny
organisms which
can only be seen
with a microscope.

In this chapter, you will learn about:

- Virus, bacteria and fungi.
- Usefulness of microorganism in food preparation and harmfulness in causing infection.
- The transmission of microorganisms and protection from these microorganisms.

All the students will be able to:

- ✓ Define microorganisms.
- ✓ Identify main groups of microorganisms and give examples of each.
- ✓ Describe the advantages and disadvantages of microorganisms in daily life.
- ✓ Define infection.
- ✓ Identify ways in which microorganisms can enter the human body.
- ✓ Suggest ways to avoid infection.

Virus, Bacteria and Fungi

Microorganisms

 Define microorganisms

Are you aware that there are millions of organisms on the planet earth? Some of them are extremely small in size and cannot be seen with the naked eye. These are called **microorganisms**. (micro - extremely small, organism - living being).

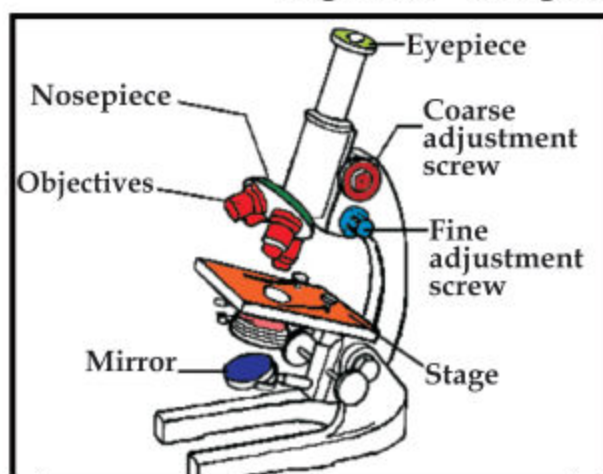


Figure 2.2: A microscope

A microscope makes things look bigger in size. Therefore we can say that a microscope magnifies objects.

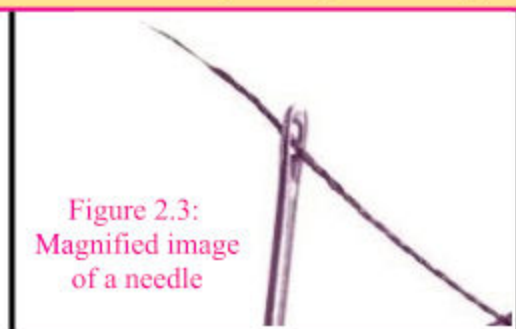
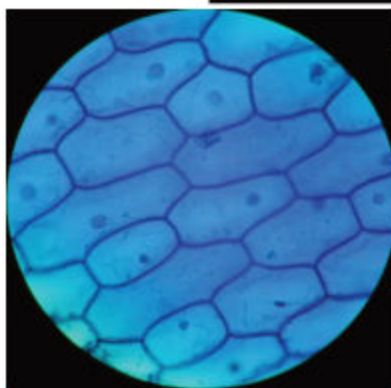


Figure 2.3:
Magnified image
of a needle

Microorganisms are present in the soil, air, water, on the surface of plants, animals and humans, and even inside their bodies.

Another name for microorganisms is microbes. They can only be seen with the help of an instrument called the microscope. Microorganisms are so small that millions of them can accommodate themselves in the eye of a needle.



Do you know that living organisms are made up of bricks called cells. Cell is the basic unit of living things. Cells are also microscopic. This picture shows cells in onion skin as seen under a microscope.

Figure 2.4: Magnified onion skin cell

Activity 1: How does a microscope work?

Collect the following materials: clips of newspaper, scissors, glue, plain paper, glass, tape, water, and ruler.

1. Cut the letters **a** and **e** from a newspaper.



2. Stick them on a piece of plain paper



3. Measure the size of the letters record in your note book .

4. Now take a glass. Stick the paper with plain side out(you can see the letters from the other side)



5 Fill the glass with water.



6. NOW SEE THE LETTERS. They are big! Magnified! Measure their size now. Record in the table below.

Letters	A Initial size (mm)	B Size with water and glass (mm)	B – A (B minus A)	C Size increase (mm)
a				
e				

Teacher Note: Engage the students in performing the activity. Encourage group work. Help taking measurements.

Major groups of microorganism: Virus, Bacteria and Fungi

Identify the main groups of microorganisms

Microorganisms are **classified** into many groups. These groups are made based on the **similarities** and **differences** among them. Microorganisms are grouped as **viruses, bacteria and fungi**.



Do you know that large organisms have trillions of cells but most microorganisms have only one cell in their bodies?

Bacteria are **single** celled micro-organisms. Bacteria feed, grow, breathe, move, and divide to produce more bacteria. This is how you may see bacteria under a microscope. Some bacteria are shaped like rods; others are round or spiral in shape. E.g a bacterium which lives in the intestines of human beings and animals is *Escherichia coli*

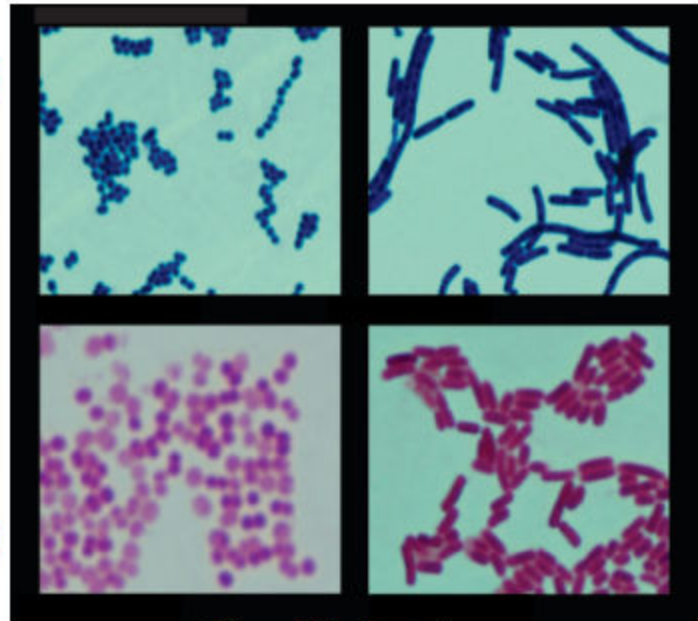


Figure 2.5: Bacteria

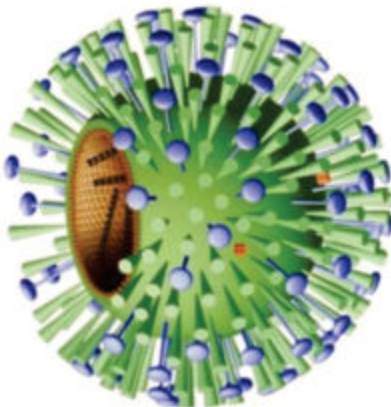


Figure 2.6: Virus

Viruses: are the simplest of all microorganisms. Scientists use high-powered microscopes to observe them. The structure of the virus is so simple; it is not even called a cell. Viruses do not feed, grow or divide on their own. They do so when they are inside other living organisms such as bacteria, animals, plants or human beings.

Do you know that a virus gives you a runny nose when you get a common cold? The common cold is caused by the **flu virus** or **influenza virus**.

Fungi: Some microorganisms belong to the group of living organisms called fungi. You should however remember that, **all fungi are not microscopic**. There are some fungi which can be easily seen with the naked eye.



Figure 2.7: Fungi

You have studied about a fungus that is not a microorganism in the previous chapter. Do you remember the fungus?

Only single celled fungi are microscopic. The cell of microscopic fungi is somewhat larger than a bacterial cell. Therefore among bacteria, virus and fungi, fungi are the largest microbes. An example of unicellular microscopic fungus is **yeast**.

Do you know that the yummy fruit bun you eat has yeast added in it?

Unscramble the given word:
suriv

Hint: A non cellular microbe

Main characteristics of bacteria, virus and fungi

Bacteria	Virus	Fungi
Unicellular organisms; cells are simpler.	Non cellular organism	Unicellular organisms; cells are more complex than those of bacteria.
Feed, grow , breathe, move and divide in order to produce more bacteria.	Grow and divide, only when they are inside other living organisms.	Feed, grow, breathe, move and divide into new fungi on their own.
	Smallest microbes	Largest microbes

Usefulness and harmfulness of microorganisms

T Describe the advantages and disadvantages of microorganisms in daily life.

Microorganisms are very important for maintaining the life on earth. These tiny organisms created by nature can be both useful and harmful. They carry out many processes in nature.

You know all of us are not bad. Some of us are very useful for you.

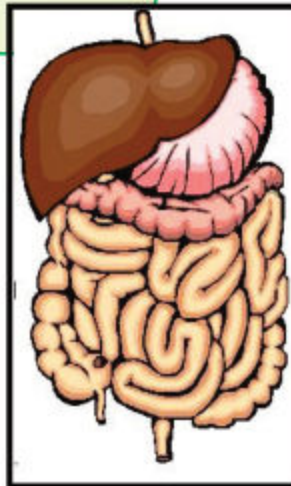


Decomposition: The bacteria in soil break down garbage and remains of living organisms into simple nutrients. Without them we would be living in huge heaps of garbage. The nutrients released into the soil, make it fertile and helpful for plant growth.



Figure 2.8: Organic waste

In Food Making: Yeast is used for the raising of bread. Some bacteria are used in the making of yogurt, cheese and sour cream, etc.



Food Digestion: Some bacteria and fungi are normally present on the surface and inside the bodies of plants, animals and humans. They help in their body processes. Cattle, goat, sheep, and rabbits have special bacteria living in their digestive tube, which help in the digestion of food.



Figure 2.9: Uses of bacteria in making food

Activity 2: Using yeast to raise bread dough and converting milk into yogurt.

What do you need

- All purpose flour
- Yeast available at a local grocery or sweet shop
- Water
- Two tablespoons of sugar
- Milk
- Yogurt starter



Figure 2.10: Dough before and after raising.

What to do?

For bread dough

1. Your teacher will mix yeast with a little luke warm water.
2. Pour this mixture into flour containing two tablespoons of sugar. Knead the mixture to make dough. Take a container, put the dough in it and level the surface. Measure the height of the dough, this height is A
3. Leave the dough at a warm place for 3-4 hours.
4. After 4 hours, measure the height of the dough again. This height is B, find B-A. Has the dough risen? How much?

Initial height of dough (cms)	Final height of dough (cms)

For yogurt

1. Heat pre-boiled milk until its Luke warm.
2. Add a table spoon of yogurt. Mix well.
3. Cover the container with a lid. Leave it undisturbed for four to five hours
4. Check the contents after five hours. What do they look like?

What did you observe?

What happens to the milk?

Bacteria present in the yogurt start changing the milk into yogurt. In a few hours, all the milk is completely converted to yogurt.

What happens to the dough?

Yeast feeds on sugar in the dough and produces carbondioxide gas.

As carbondioxide is released in the dough, it rises up.

Activity questions:

Do you think that the dough will rise if the yeast is put in ice cold water?

Do you think that the dough will rise if the yeast is put in boiling water?

In antibiotics and vaccines:

When did your doctor prescribe you an antibiotic? Some microbes produce chemicals called **antibiotics**. These chemicals destroy living cells, particularly micro organisms. They are used for the treatment of some diseases. **Penicilium** is a fungus that produces antibiotic called **penicillin**.

Microbes also produce **vaccines**. Vaccines prevent us from contracting certain diseases. Polio drops are a form of vaccine, made from the polio virus.



Figure 2.11: Doctor giving Polio drops to a child

Unscramble the given word:
ntiaoibtc

Hint: A chemical that destroys living cells

Teacher Note: Perform activity in class. Encourage students to observe the changes in dough and take measurements. Let them taste the yogurt and notice that the milk has changed.



Figure 2.12:Detergent powder

They are also used in the making of **leather** and **detergent powders**.



Figure 2.13: A pair of leather shoes

Harmful effects of microorganism



Figure 2.14: A rotten orange

Spoiling of food:

Some harmful bacteria get into our food and produce harmful chemicals. The food goes bad and produces a foul smell. Vegetables and fruits also rot when certain microbes attack them.

Contamination of drinking water:

Do you know many harmful microorganisms live in water? They make the water unfit for drinking. This water can only be purified from these microbes, by boiling it for at least five minutes. Boiling kills microbes.



Figure 2.15: A rotten cucumber

Activity 3: Observe the spoilage of food; growth of a fungal colony on a piece of bread.

A colony is a group of same kind of individuals living together.

What do you need?

A piece of moist bread.

What to do?

Take a piece of bread or roti. Moisten it and keep it at a warm open place. You may keep it in an open jar with a little water at the base.

Observe the bread piece every day. What changes do you observe? Do you see anything forming on the piece of bread? Use a hand lens to observe.

Moisten the piece of bread or roti if it gets dry.

What did I observe?

Days	What's new in my jar today?

Activity questions:

What is the colour of the colony?

Does the size of bread piece or roti change with the growth of colony?

Infections and diseases



Define infection

How often does your doctor tell you that you have an infection? What is an infection? **Any disease caused by the presence, and increasing multiplication of microbial organisms in the body, is called an infection or infectious disease.** Common cold or flu in which a person has a runny nose, sneezing, fever and headache is a viral infection.

Teacher Note: Engage the students in performing the activity. Observe the changes every day. Help them look at the structure of the fungal colony with a hand lens.

Some kinds of diarrhoea are caused by a bacterial infection. Various skin problems such as rashes, etc, are fungal infections. Plants and animals are also infected by various microbes.



Figure 2.16: A boy down with flu and fever; a viral infection



Figure 2.17: Athlete's foot: fungal infection of the foot

Routes to infection

Identify ways in which microorganisms can enter the human body

You know that microbes are present everywhere, in air, in soil, in water bodies, public bathrooms and on various surfaces. They may enter the human body through one of the following routes:

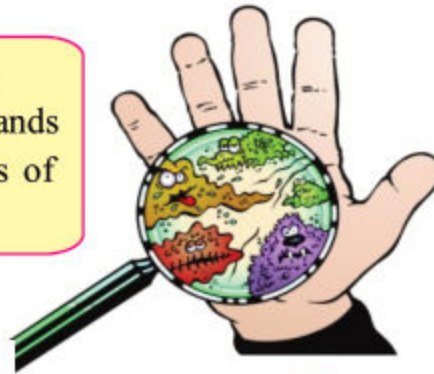
through the mouth, while eating

through the nose, while breathing

through the cuts in skin



Do you know? infection causing microbes are very good friends of our hands. Our hands touch many surfaces, so they carry germs of all kinds.



Microbe talk!



We are the bad microbes called germs. We cause typhoid, jaundice and athletes foot. Have you ever been infected by us?

Preventing infection

i Suggest ways to avoid infections

Do you know how you can protect yourself from catching

infections? The only way to stop the spread of infections is to maintain hygiene. Hygiene means cleanliness. Hygiene includes personal and environmental hygiene.



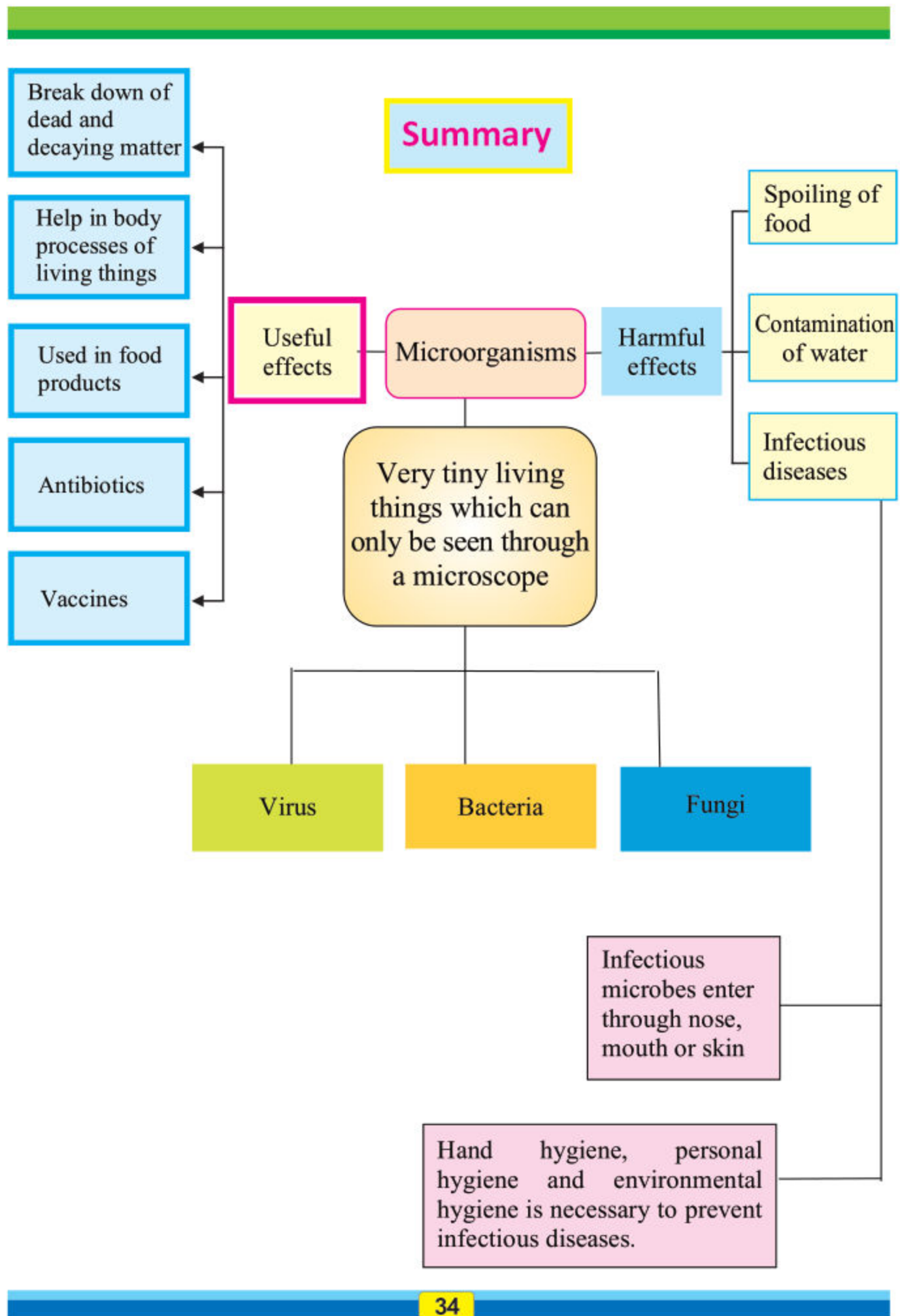
Figure 2.18: Ways to maintain hygiene

Personal hygiene includes washing hands frequently, trimming nails, bathing regularly and brushing teeth twice a day. Washing hands with soap and water after using toilet is also necessary. A clean body is free from harmful microbes that may cause diseases.

Environmental hygiene means that streets, roads, water bodies and bare grounds should be kept clean. Harmful microorganisms do not grow at clean places.

Unscramble the given word:
ctionfein

Hint: A disease caused by microorganisms



Project

Stop the spread

What do you need:

- List of infectious diseases and questions to explore, how are they caused and how do they travel from person to person.
- List of people such a doctor, nurse, dispenser, science teacher you could talk to.
- List of websites.
- Poster, chart paper and markers.

What to do: Choose a common infectious disease that affects human beings, plants or animals. Explore the following questions about it.

- Which microorganism is the cause of the disease?
 - What happens to the diseased person? (what happens to the crop plant or animal)
 - Where the microorganism is normally found?
 - How does it enter the body of plant, animal or human?
 - What can be done to stop the spread of disease?
 - Use interesting figures or puppets to show your microbe in class.
1. Discuss with your family members, teachers, pair and senior fellows to find answers to the given questions and those that you have listed.
 2. Explore books, websites, tv and radio or text your family and friends.
 3. Present your findings through a poster titled “Stop the Spread” or a story called a “Germ Story” or a role play titled “Meet us”.

Review questions:

1. Fill in the blanks:

- Yeast is an example of a _____
- E. coli is an example of _____
- Flu is caused by a _____
- An example of antibiotic is penicillin, which is produced by a fungus called _____
- An example of vaccine is _____. It works against the polio disease.

2. Mark true or false:

- All microbes cause diseases. T / F
- Microorganisms have bodies composed of just one cell. T / F
- Viruses are single celled and can live, grow and divide outside living cells. T / F
- Athlete's foot is a fungal infection T / F
- Flu is a bacterial infection T / F

3. Define and give an example of each of the following:

microorganisms

microscope

antibiotics

infection

personal hygiene

environmental hygiene

4. Answer the following questions:

- Briefly describe the three groups of microorganisms. Give examples for each.
- Make a list of good hygiene habits.
- Omer has been diagnosed with a bacterial throat infection. His sister, Abida, has a viral infection called measles. Write in your own words what each of these mean?

5. This is a gang of ugly and bad microbes. They call themselves BAD GERMS. Do you know how they harm human beings animals and plants? Write in your own words.



Describe ways in which we can save ourselves from the GERM GANG.

These are friendly microbes. Do you know how they are important for human beings and the environment?



Figure 2.17: Probiotics

6. Scientific Problem Solving

The people of the Province of Sindh, Pakistan, are facing a serious problem these days. Many people are developing a disease called Dengue. The diseased person develops high fever, skin rash and sometimes may bleed.

Doctors explain the disease as the following: Dengue is a disease caused by the bite of one kind of mosquito. The mosquito contains a virus called dengue virus. When the mosquito bites a human, it transfers the virus into the human body. The mosquito causing dengue thrives in fresh water ponds.

The community is thinking of solutions to fight dengue. Can you help in coming up with some solutions? Discuss with doctor, nurse, dispenser, science teacher, senior fellows or call family members. Share your solutions with your class fellows.

7. **Environmental Hygiene is important in order to protect ourselves from germs. Observe the home environment and share ways that germs can enter our bodies. Suggest ways to clean the environment and protect the family from germs.**



Figure 2.18: Unhealthy home environment